

Player CRC Card

Class name: Player (self, texture)	
Class Attributes:	Class Collaborations (other classes):
• <code>self.texture</code> # instance variable that holds the object's sprite • <code>self.wallet</code> # instance variable that holds the value of the amount of Coins	• Composes with PyGame • Collaborates with Coin
Class Methods:	Class Collaborations (other classes):
• <code>__init__()</code>: ◦ Creates a <code>Player</code> object that can move around the world and collect coins • <code>__str__()</code>: ◦ Makes the <code>str()</code> function work with the <code>Player</code> • <code>movement()</code>: ◦ Moves the <code>Player</code> across the tilemap per tile • <code>attack()</code>: ◦ Depletes <code>attack_power()</code> from current battling <code>bad_NPC</code>'s health	• Composes with PyGame • Collaborates with Coin
Class Variables:	Class Collaborations (other classes):
• <code>attack_power</code>: ◦ A variable that holds a set amount of damage that will be dealt to a <code>bad_NPC</code>'s health upon a hit • <code>health</code>: ◦ Variable that holds an integer amount; is depleted from when <code>bad_NPC</code> attacks	

NPC CRC Card

Class name: NPC (self, texture, is_random = False, is_key = False, key_dialogue = None, region = None)

Class Attributes:

- `self.texture` # instance variable that sets the object's sprite to the `sprite` parameter
- `Self.is_random = False` # instance Boolean that declares whether the NPC will remain stationary or move randomly within their region
- `Self.is_key = False` # instance Boolean that declares whether the NPC is a key NPC (unique dialogue/action; boss or important person) or a default NPC (draws dialogue randomly from a set list)
- `self.key_dialogue = None` # optional (only for those meant to be key NPC's); instance variable that is set to the `key_dialogue` parameter to be unique for every instance of a key NPC object
- `self.region = None` # optional (only for NPC's that do not remain stationary); instance variable that is set to the `region` parameter that declares where the NPC will idly walk around at

Class Collaborations (other classes):

- Composes with `pygame`
- Composes with `Player`

Class Methods:

- `__init__()`:
 - Creates an NPC object that interacts with the player and may move idly; can be Key or not (provides unique dialogue); can be stationary or not (has an associated region in which to wander)
- `__str__()`:
 - Makes the `str()` function work with the NPCs
- `movement()`:
 - If the current NPC has an associated region, have them move per tile within the boundaries of it; else, have them stay still

Class Collaborations (other classes):

- Composes with `pygame`
- Composes with `Player`

Class Variables:

- `default_dialogue`:
 - List of pre-determined dialogue options for NPC's to display upon interaction if `is_key` returns False

Class Collaborations (other classes):

Bad NPC CRC Card

Class name: bad_NPC (self, texture)	
Class Attributes:	Class Collaborations (other classes):
• <code>self.texture</code> # instance variable that sets the object's sprite to the sprite parameter	• Inherits from NPC
Class Methods:	Class Collaborations (other classes):
• <code>__init__()</code>: ◦ Creates an NPC object with the additions of health and attack_power • <code>__str__()</code>: ◦ Makes the <code>str()</code> function work with the badNPC • <code>attack()</code>: ◦ Depletes attack_power from Player's health • <code>initiate_battle()</code>: ◦ (like <code>interaction()</code>) If the current NPC is a key NPC, display their unique dialogue; else, draw randomly from <code>default_dialogue</code>; loop <code>attack()</code> back and forth between Player and bad_NPC; if bad_NPC health reaches 0, <code>collect()</code>	• Inherits from NPC • Collaborates with Coin
Class Variables:	Class Collaborations (other classes):
• <code>attack_power</code>: ◦ A variable that holds a set amount of damage that will be dealt to the Player's health upon a hit • <code>health</code>: ◦ Variable that holds an integer amount; is depleted from when bad_NPC attacks	

Good NPC CRC Card

Class name: good_NPC (self)	
Class Attributes:	Class Collaborations (other classes):
• <code>self.texture</code> # instance variable that sets the object's sprite to the sprite parameter • <code>self.has_interacted</code> # instance Boolean that declares whether the Player has interacted with the good_NPC before	• Inherits from NPC
Class Methods:	Class Collaborations (other classes):
• <code>__init__()</code>: ◦ Creates an NPC object with the additions of health and attack_power • <code>__str__()</code>: ◦ Makes the <code>str()</code> function work with the good NPC • <code>interaction()</code>: ◦ Sets <code>has_interacted</code> to True and calls <code>collect()</code>; if the current NPC is a key NPC, display their unique dialogue; else, draw randomly from <code>default_dialogue</code>	• Inherits from NPC • Collaborates with Coin
Class Variables:	Class Collaborations (other classes):

Environment CRC Card

Class name: Environment (self)	
Class Attributes:	Class Collaborations (other classes):
• <code>self.texture</code> # instance variable that sets the object's sprite to the sprite parameter	• Collaborates with PyGame • Collaborates with Player • Collaborates with NPC
Class Methods:	Class Collaborations (other classes):
• <code>__init__()</code>: ◦ Sets up the current environment in which the player will explore • <code>__str__()</code>: ◦ Makes the <code>str()</code> function work with the Environment • <code>setup_start_menu()</code>: ◦ Sets up the start screen, where the user can start or exit the game. • <code>setup_pause()</code>: ◦ Sets up the pause/menu screen, in which the user can continue/exit the game • <code>setup_spawn()</code>: ◦ Set up the spawn point for the player • <code>setup_town()</code>: ◦ Set up the environment, Town • <code>setup_shop()</code>: ◦ Set up the environment, Shop	• Collaborates with PyGame • Collaborates with Player • Collaborates with NPC
Class Variables:	Class Collaborations (other classes):
• <code>environment_name</code>: ◦ A list containing the different environments	

Obstacle CRC Card

Class name: Obstacle (self, texture, is_interactable = False)	
Class Attributes:	Class Collaborations (other classes):
• <code>self.texture</code> # instance variable that sets the object's sprite to the <code>sprite</code> parameter • <code>self.is_interactable</code> # instance Boolean that declares whether the object will present the option to interact or not • <code>self.dialogue</code> # instance variable that holds what the Obstacle says upon <code>interaction()</code>	• Collaborates with PyGame • Collaborates with Player • Collaborates with <code>good_NPC</code> • Collaborates with <code>bad_NPC</code>
Class Methods:	Class Collaborations (other classes):
• <code>__init__()</code>: ◦ Sets up the different attributes of the obstacle being created • <code>__str__()</code>: ◦ Makes the <code>str()</code> function work with the different obstacles • <code>interaction()</code>: ◦ If <code>is_interactable</code> is True, allows player the option to view dialogue from the Obstacle	• Collaborates with PyGame • Collaborates with Player • Collaborates with <code>good_NPC</code> • Collaborates with <code>bad_NPC</code>
Class Variables:	Class Collaborations (other classes):

Coin CRC Card

Class name: Coin (self)	
Class Attributes:	Class Collaborations (other classes):
• <code>self.texture</code> # instance variable that sets the object's sprite to the sprite parameter • <code>self.amount</code> # the current amount of coins	• Collaborates with PyGame • Collaborates with Environment • Collaborates with Player
Class Methods:	Class Collaborations (other classes):
• <code>__init__()</code>: ◦ Creates the currency that will be used inside of the game • <code>__str__()</code>: ◦ Makes the <code>str()</code> function work with the Coin • <code>collect()</code>: ◦ Randomly adds to the amount of Coins to the player's wallet attribute.	• Collaborates with PyGame • Collaborates with Environment • Collaborates with Player
Class Variables:	Class Collaborations (other classes):

Shop CRC Card

Class name: Shop (self)	
Class Attributes:	Class Collaborations (other classes):
• <code>self.texture</code> # instance variable that sets the object's sprite to the sprite parameter	• Inherits from Environment • Collaborates with PyGame • Collaborates with Coin • Collaborates with Player
Class Methods:	Class Collaborations (other classes):
• <code>__init__()</code>: ◦ sets up the shop menu • <code>__str__()</code>: ◦ Makes the <code>str()</code> function work with the shop • <code>buy()</code>: ◦ Helps the <code>Player</code> buy collectibles from the shop • <code>sell()</code>: ◦ Helps the <code>Player</code> sell collectibles from the shop	• Inherits from Environment • Collaborates with PyGame • Collaborates with Coin • Collaborates with Player
Class Variables:	Class Collaborations (other classes):

Game CRC Card

Class name: Game (self)	
Class Attributes:	Class Collaborations (other classes):
• <code>self.size</code> # initializes the screen size • <code>self.running</code> # initializes whether or not the game is running • <code>self.screen</code> #initializes the screen by using <code>pygame.display</code> • <code>self.environment</code> # initializes the environment by using the Environment class • <code>self.clock</code> # initializes a clock object that keeps track of time • <code>self.player</code> # initializes the player object	• Collaborates with PyGame • Collaborates with Player • Collaborates with good_NPC • Collaborates with bad_NPC • Collaborates with Shop • Collaborates with Environment • Collaborates with Obstacle
Class Methods:	Class Collaborations (other classes):
• <code>__init__()</code>: ◦ Sets up all the necessary information to build the game canvas and logic • <code>__str__()</code>: ◦ Makes the <code>str()</code> function work with the Game • <code>run()</code>: ◦ Runs the game itself with the collaboration of all the aforementioned classes.	• Collaborates with PyGame • Collaborates with Player • Collaborates with good_NPC • Collaborates with bad_NPC • Collaborates with Shop • Collaborates with Environment • Collaborates with Obstacle
Class Variables:	Class Collaborations (other classes):